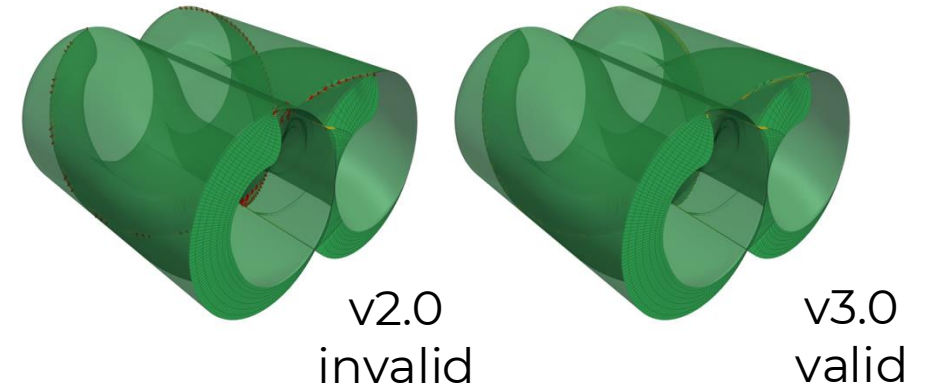
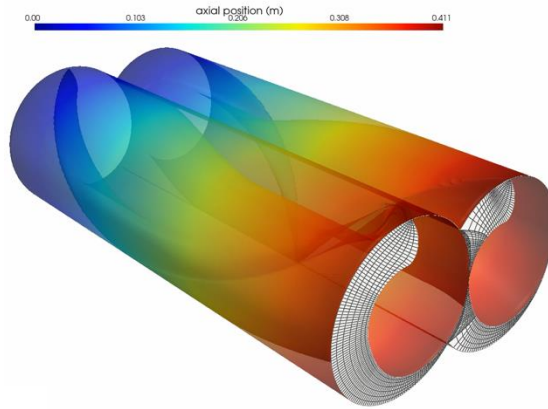




SplineMesh v3.0

**Robust spline-based structured mesh generation for
positive-displacement rotary machines with sharp features**

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Ye Ji • Matthias Möller

Delft Institute of Applied Mathematics, Delft University of Technology

Paper 1039 • y.ji-1@tudelft.nl



Outline

- 1. Background and motivation**
- 2. The v3.0 robustness layer**
- 3. Results and runtime cost**
- 4. Conclusions and outlook**

Robust meshing is essential for reliable CFD

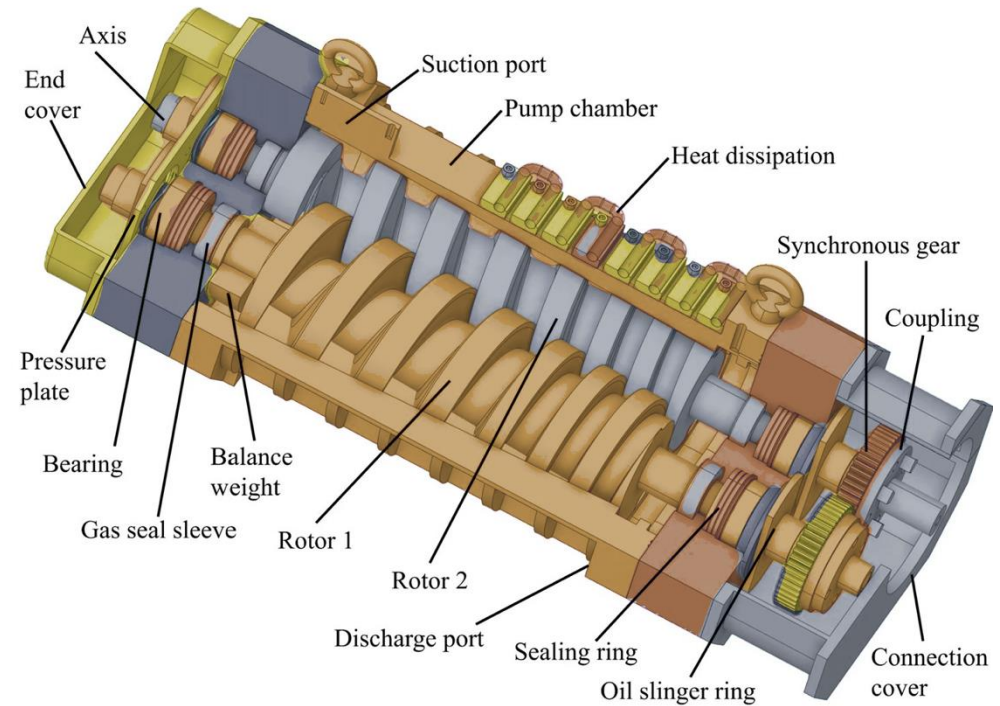
- Computational fluid dynamics (CFD) is central to the design and performance prediction of **positive-displacement rotary machines**, e.g., screw compressors, expanders, blowers, and dry vacuum pumps;
 - **Volumetric efficiency** is governed by sealing gaps of only tens of μm on ~ 100 mm rotors (scale ratio $\sim 10^{-4}$);
 - **Reliable CFD requires high-quality meshes** throughout the continuously deforming clearances between intermeshing rotors.
- **A single folded element can cause the downstream CFD simulation to fail.**

SplineMesh — spline meshing on G+Smo + SCORG™

v1.0 methodology

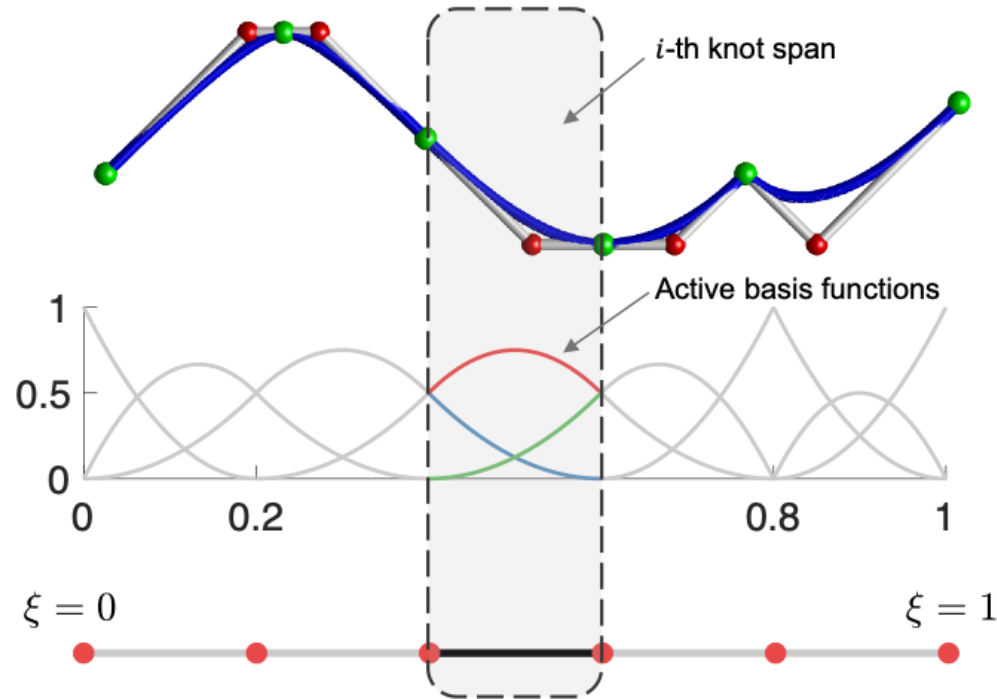
v2.0 efficiency

v3.0 robustness



Twin-screw dry vacuum pump. The computational domain is the gas volume between the rotors and casing.

We mesh the exact spline geometry, not a polyline approximation

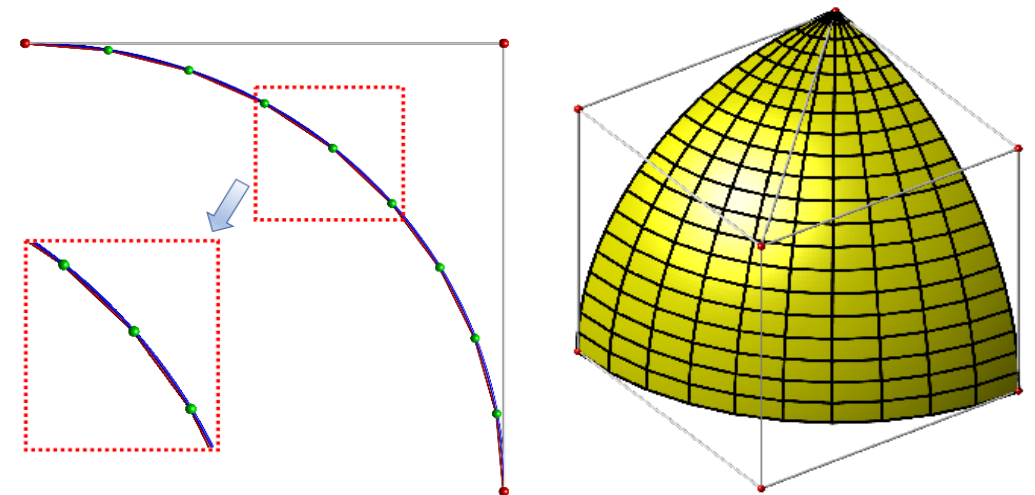


- Conventional meshing relies on a piecewise-linear approximation, while **SplineMesh works directly with the spline geometry**;
- **Exact boundary**: the mesh conforms to the true profile;
- **Compact**: far fewer geometric control points;
- **Unified representation** for arcs, cycloids, free-forms.

Piece-wise (rational) polynomial description

$$\mathcal{C}(\xi) = \sum_{i=0}^n \mathbf{P}_i \frac{\omega_i N_{i,p}(\xi)}{\sum_{i=0}^n \omega_i N_{i,p}(\xi)}, \xi \in \hat{\Omega},$$

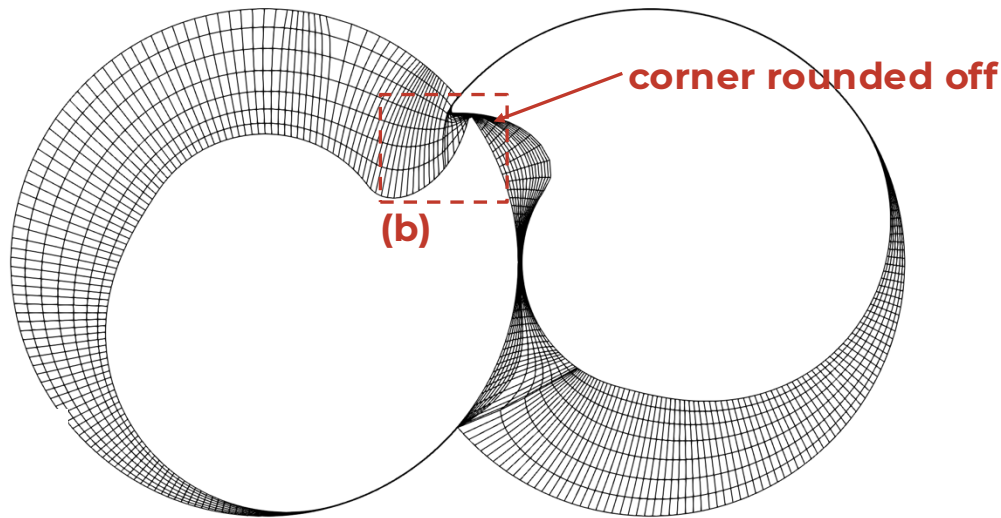
$$\text{with } N_{i,p}(\xi) = \frac{\xi - \xi_i}{\xi_{i+p} - \xi_i} N_{i,p-1}(\xi) + \frac{\xi_{i+p+1} - \xi}{\xi_{i+p+1} - \xi_{i+1}} N_{i+1,p-1}(\xi)$$



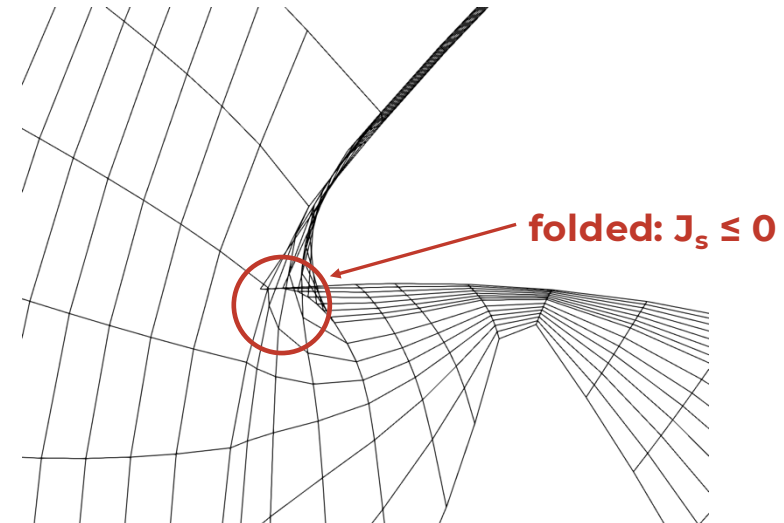
3 control points vs.
10 grid points

9 control points vs.
231 grid points

Vacuum-pump geometry exposes two critical v2.0 blockers



(a) cusp rounded off the lobe tip



(b) boxed region in (a), magnified

① Geometric issue: unconstrained smoothing rounds off the sealing corner

- The smoother pulls nodes away from the tip, rounding off the sharp sealing feature;
- Thin strips adjacent to the tip develop **inverted elements** (scaled Jacobian $J_s \leq 0$).

② Licensing issue: v2.0 cannot be commercially deployed

Boundary matching relied on a triangulator whose licence is **incompatible with commercial deployment**.

How existing meshing approaches handle invalid elements

Approach	Geometry representation	Clearance resolution	Invalid-element handling
Algebraic (SCORG™) / Laplacian ALE / TwinMesh	piecewise linear	resolved	No explicit repair
Fictitious domain	background grid	not body-fitted	n/a
Unstructured (Gmsh)	simplicial	clearance resolved, but unstructured	remeshing
SplineMesh v2.0	spline	resolved	none
SplineMesh v3.0 (this work)	spline	resolved	detect + untangle

- To our knowledge, no published method provides an **explicit detect-and-repair mechanism** for this class of structured clearance meshes;
 - In practice, failures often require re-blocking or relaxed clearance constraints.
- We therefore introduce a **robustness layer** applied to each completed quadrilateral slice mesh;
 - It can, in principle, be **integrated into other mesh-generation pipelines**.

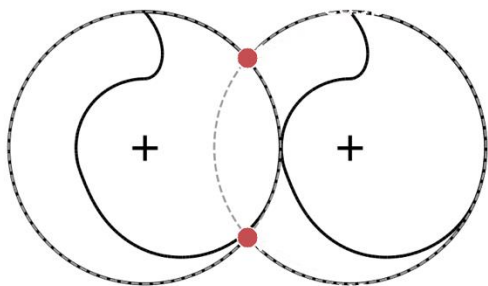


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A four-stage workflow from geometry to valid mesh

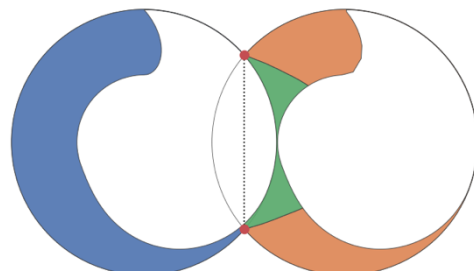
existing pipeline



(a) Identify cusps

- Input rotor profiles + casing;
- Cusps fix the corners

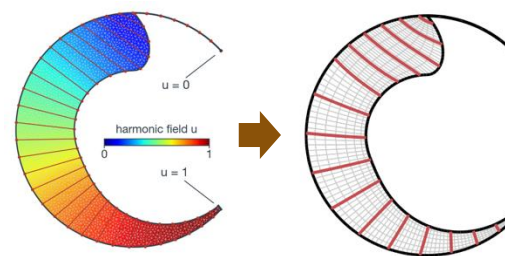
existing pipeline



(b) Partition into blocks

- Split the domain at the cusps into male, female, and inter-lobe blocks

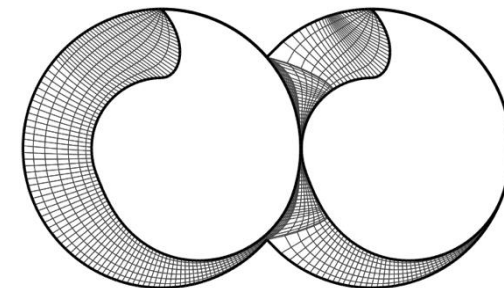
new in v3.0



(c) Match boundaries

- Harmonic field establishes node correspondence along radial lines

new in v3.0



(d) Smooth + untangle

- Elliptic smoothing followed by local untangling of folded elements

SplineMesh v3.0: a corner-preserving detect-and-repair layer

- Sharp sealing corners must be preserved.
- v2.0 smoothing can round them off and trigger nearby element folding.

v3.0 preserves the corner and repairs the surrounding mesh.

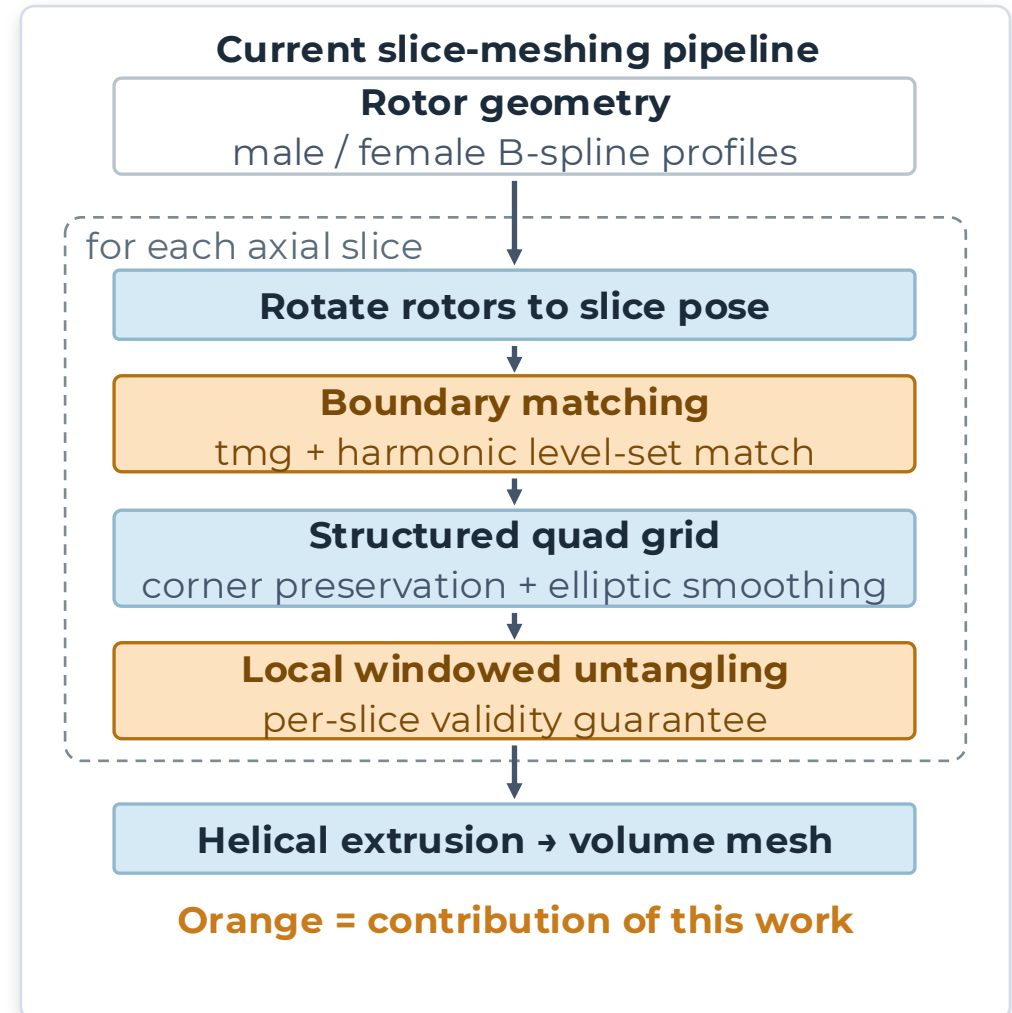
Two new components:

1 • Deployment-compatible triangulator (tmg)

- Replaces Triangle in boundary matching, removing the deployment blocker.
- No additional third-party licensing restriction is introduced beyond the existing G+Smo dependency.

2 • Local windowed untangling

- Triggered only when invalid elements are detected.
- Repairs folded cells locally while preserving the sealing corner.



Deployment-ready triangulator (tmg)

Where it is used

Boundary matching uses a discrete harmonic solve on an auxiliary **triangulation** of each sub-region.

Was — Jonathan Shewchuk's Triangle Library

Excellent quality, but its licence is **incompatible with commercial deployment**.

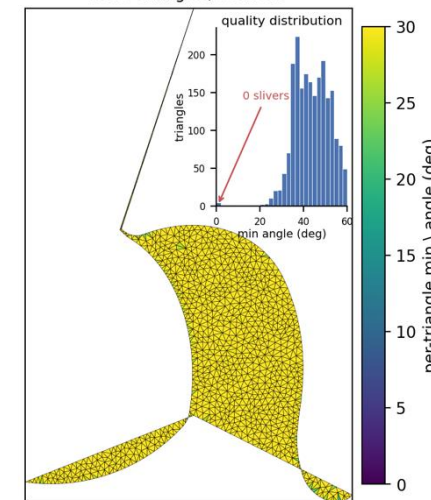
Now — tmg, built in

Constrained Delaunay with **exact predicates**; header-only, no external dependency.

Robust output

No near-degenerate triangles, a stable auxiliary mesh for the harmonic solve

tmg triangulation of a thin clearance boundary
2105 triangles, 0 slivers



Thin clearance boundary: dense, graded, zero slivers.

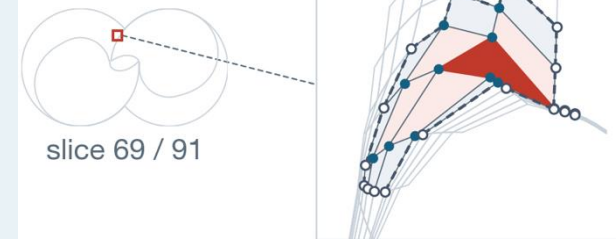
4–10% end-to-end cost

Same triangle count (within 1.6%); $\approx 4\text{--}5\times$ per tmg call.

Local windowed untangling

- **Metric: scaled Jacobian** J_s minimum signed corner Jacobian;
 - $J_s \leq 0$ **means folded**;
- **Window**: the k -ring neighborhood around each invalid cell.

k = 2 window



Two stages: the barrier-type energy requires a valid initialization, so validity must be recovered first.

Stage 1: open the folds

- **graph-Tutte energy + as-rigid-as-possible term**
- optimized with vector-Adam opt. solver; Newton-type iterations are unreliable from invalid starts



Stage 2: improve the mesh quality

- target-shifted barrier, default $J_s^* = 0.05$
- **updates accepted only if validity is preserved and mesh quality improves**

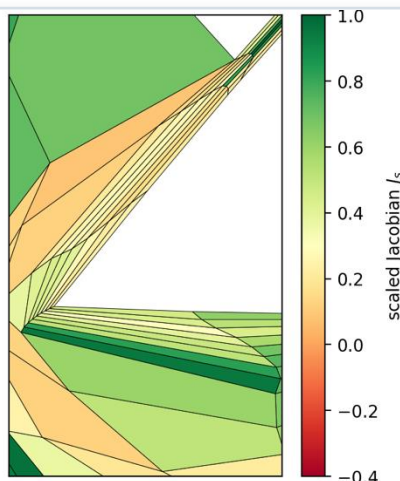
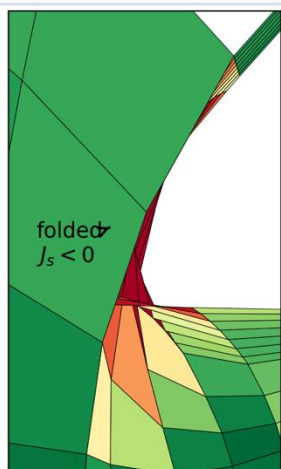
- **QUIMBY stack: 271–530 folded quads → 0.**
- Clean slices are skipped, thus computational cost is incurred only where repair is needed.
- No machine-specific algorithm parameter tuning.



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Robust untangling across three rotary-machine families



- Left: before, **23 folded quads** (red); right: after, **0 folds**.
- Boundary nodes unchanged.

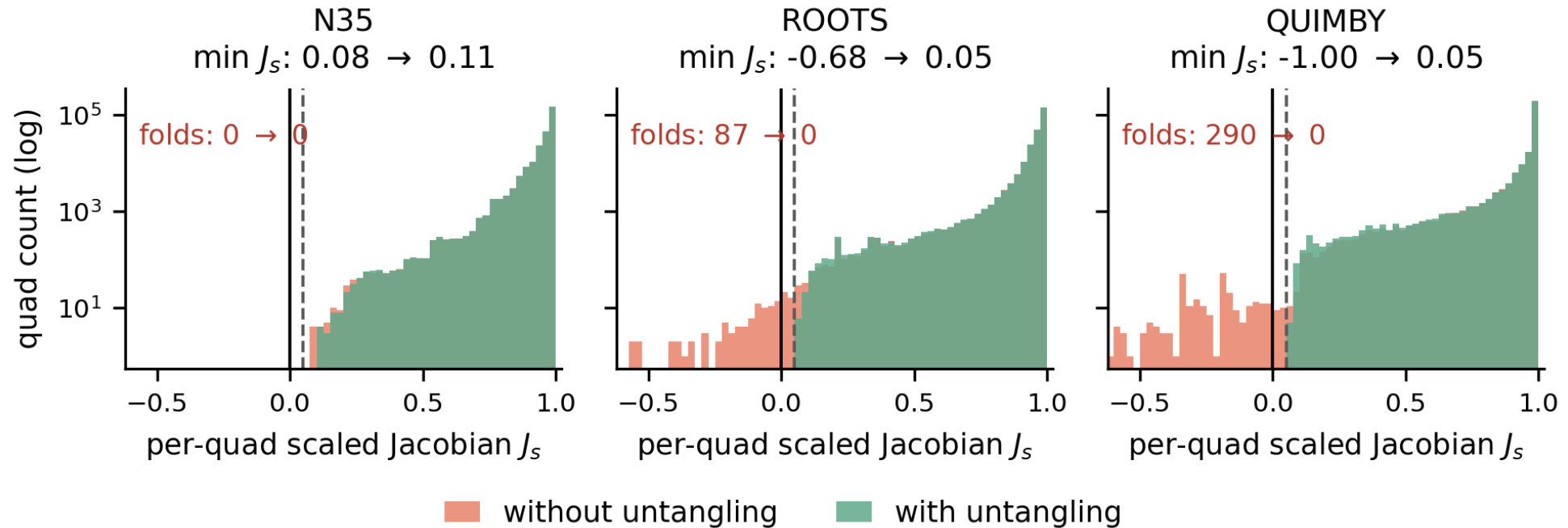
≈ 25 ms

per repaired slice

Slices above the quality threshold are **detected automatically and skipped**.

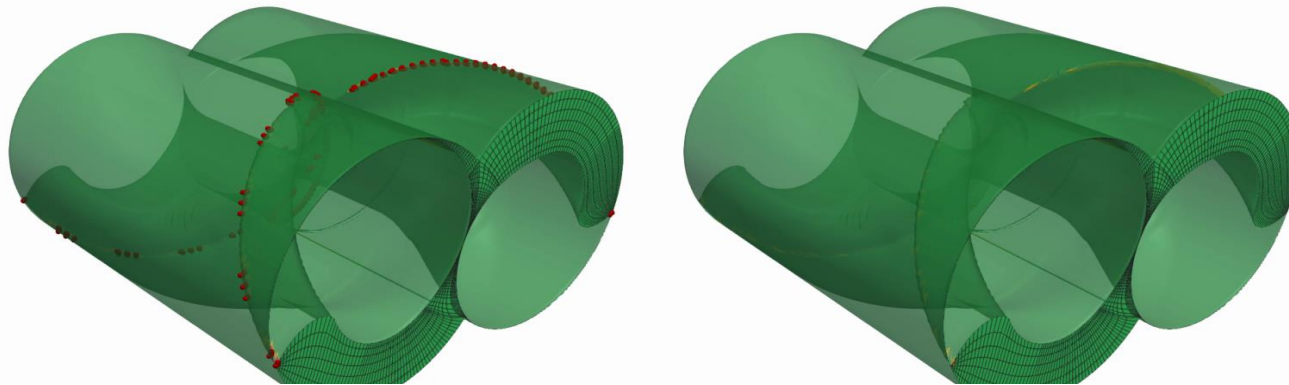
case	slices flagged ($J_s < 0.05$)	min J_s , before	min J_s , after	folded quads, before	folded quads, after	untangling time (% total)
N35 (compressor)	4	0.079	0.203	0	0	0.08 s (5.8%)
ROOTS (blower)	39	-0.682	0.055	87	0	1.10 s (48.0%)
QUIMBY (vacuum pump)	85	-1.000	0.051	290	0	2.13 s (63.6%)

Untangling removes the invalid tail and improves the mesh quality

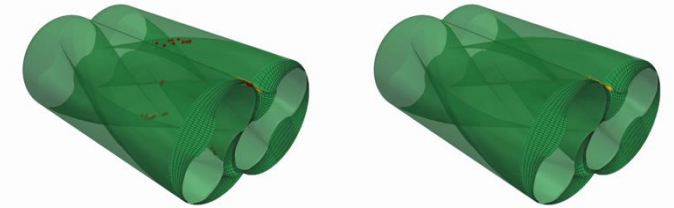


- Per-quad J_s over all 91 slices, untangling off vs on. Solid line: **validity ($J_s = 0$)**; dashed: **target $J_s^* = 0.05$** ;
- **N35 compressor is already valid; untangling only improves its mesh quality;**
- **ROOTS and QUIMBY: the entire invalid tail is eliminated.**

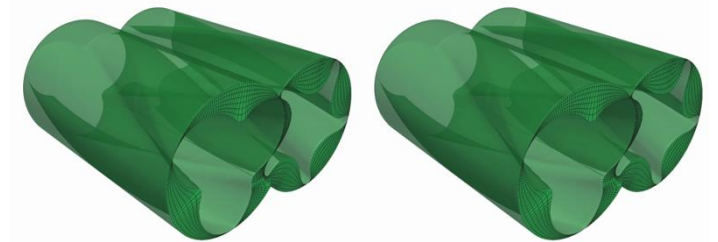
Untangling removes the inverted cells



- QUIMBY vacuum pump, full volume at true axial scale
- Left, off: **82 / 91 slices have inverted cells** Right, on: **0 inverted cells**
- Scaled Jacobian: green = good, yellow = low quality, red = inverted



ROOTS (blower)



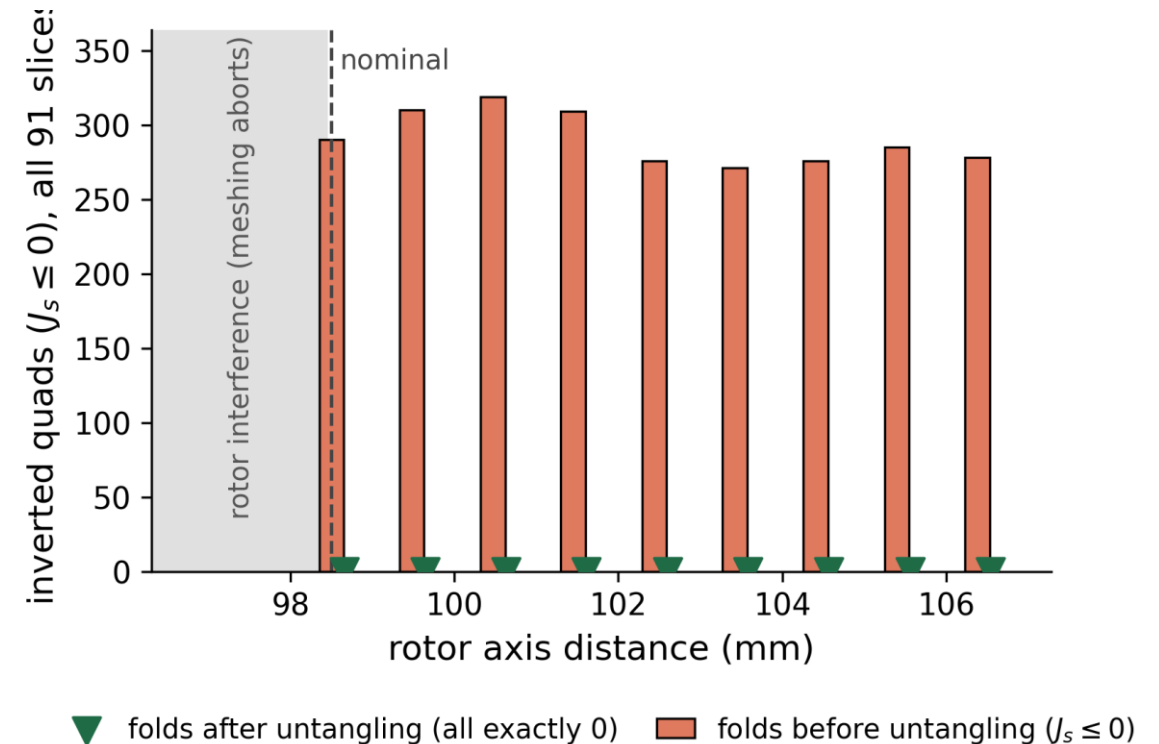
N35 (compressor)

44 of 45 configurations mesh the full cycle

	v2.0	v3.0
design clearance (9)	0/9	9/9
full parameter grid (45)	0/45	44/45

The configuration sweep

- 9 centre distances × 5 clearance scales ×
91 axial positions = **4095 slice meshes**
- centre distance 98.50–106.38 mm;
 - clearance scale 0.70–1.30.



LIVE DEMO



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The full cycle now meshes, and is ready for deployment

Conclusions

- SplineMesh v3.0 introduces **automatic fold detection and local untangling** — corner-preserving, local, and requiring no machine-specific algorithm parameter tuning;
- The complete QUIMBY cycle meshes successfully across **all 44 geometrically feasible configurations**, including all design-clearance cases;
- A **built-in triangulation backend removes** the final **deployment barrier** for integration into SCORG™.

Key results

Full-cycle success (design)

0% → 100%

Off-design grid (45)

100% (44/44)

1/45 geometrically infeasible

min J_s

-1.000 → ≥ 0.05

Licence blocker

removed

Future work

- 2D slice validity is established; 3D extrusion validity remains to be quantified per cell.
- CFD-level validation, including leakage prediction, is the next step.

Many thanks for your kind attention!

Q&A

If you are interested in my research, please feel free to contact me! ;-)

-  Email: y.ji-1@tudelft.nl
-  GitHub: [jiyess](https://jiyess.github.io/)
-  Homepage: <https://jiyess.github.io/>